Exponentials & Polynomials

Laws of exponents

1 Simplify $x^5 \cdot x^3$.

2 Simplify $\dfrac{y^9}{y^4}$.

3 Simplify $(2x^3)^4$.

Negative and fractional exponents

4 Simplify x^{-3}, writing your answer without a negative exponent.

5 Simplify $16^{3/4}$.

Adding, subtracting, and multiplying polynomials

6 Simplify $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$.

7 Expand $(x + 3)(x - 5)$.

8 Expand $(2x - 1)^2$.

Factoring polynomials

9 Factor $x^2 - 16$ completely.

10 Factor $2x^2 + 8x$ completely.

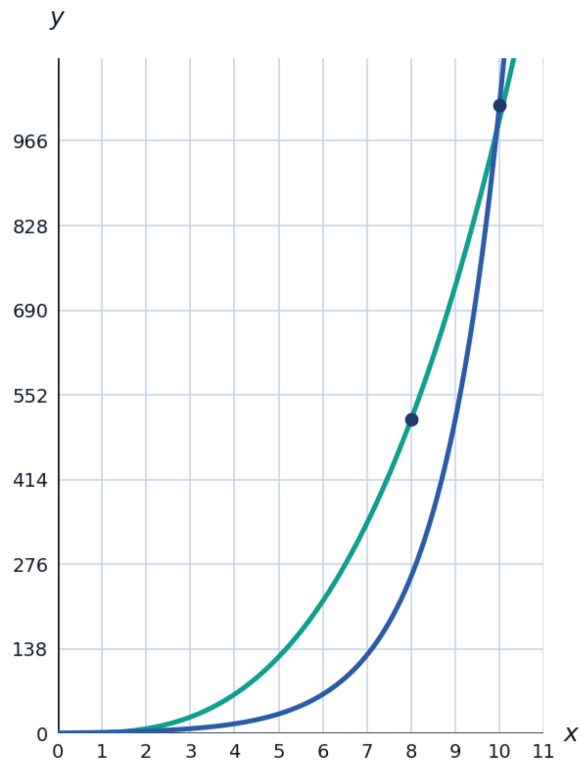
11 Factor $x^3 - 4x$ completely.

Polynomial division and the remainder theorem

12 Find the remainder when $f(x) = x^3 - 2x^2 + x - 5$ is divided by $(x - 2)$, using the remainder theorem.

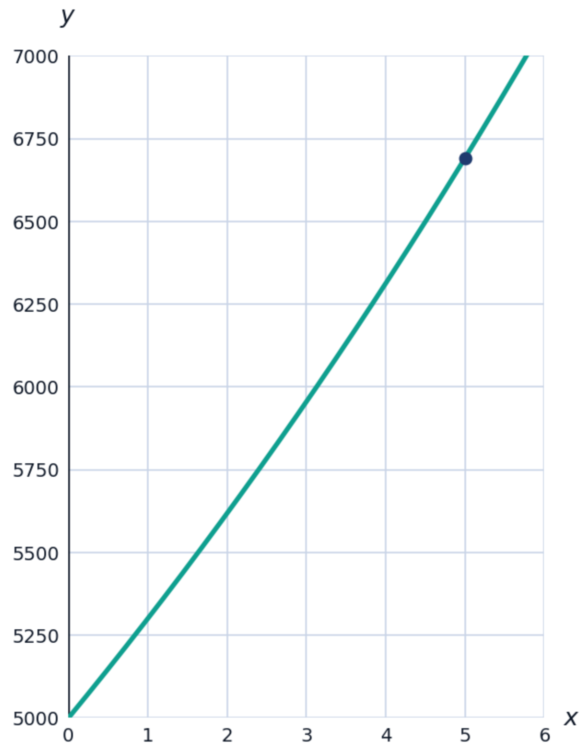
Exponential growth vs. polynomial growth

13 Two functions are given: $f(x) = x^3$ and $g(x) = 2^x$. Explain which function eventually grows faster as x increases, and estimate the crossover point using the values $f(10) = 1000$, $g(10) = 1024$, $f(8) = 512$, $g(8) = 256$.



Modeling with exponential functions

14 An investment of 5000 dollars grows according to $A(t) = 5000(1.06)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.



Applying the zero product property to polynomial equations

15 Solve $x^3 - 5x^2 + 6x = 0$ by factoring completely and applying the zero product property.

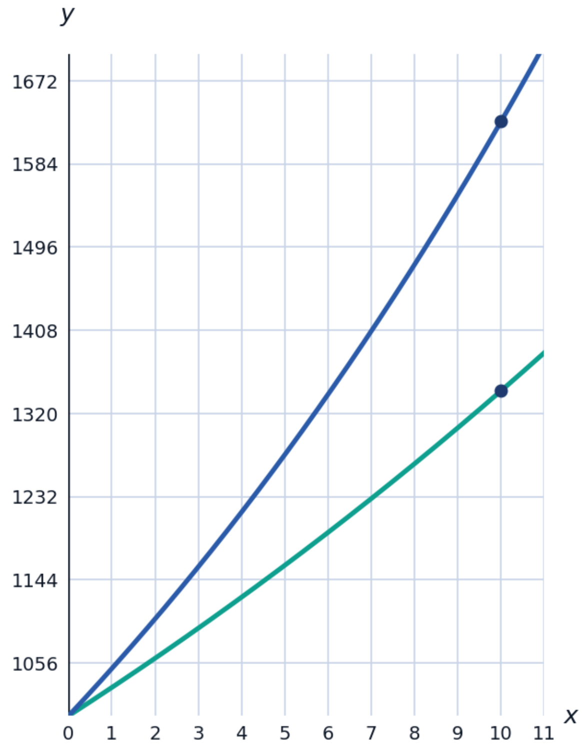
Multiplying binomials and trinomials

16 Expand $(x + 2)(x^2 - 3x + 4)$.

17 Expand $(2x - 3)(x + 5)$.

Comparing exponential models

18 Two savings accounts grow according to $A(t) = 1000(1.03)^t$ and $B(t) = 1000(1.05)^t$. Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.



Simplifying expressions with multiple exponent rules

19 Simplify $\dfrac{(x^3y^2)^2}{x^4y}$.

20 Simplify $\left(\dfrac{2x^2}{y}\right)^3$.

Half-life and decay applications

21 A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after t years, and find the amount remaining after 24 years.